

DAY – 12

DATA EXPANSION AND CLEANING

In this task, I focused on dataset expansion and cleaning to improve the performance and reliability of the Virtual Try-On system. Since AI-based try-on systems depend heavily on image quality and data variety, collecting and organizing a structured dataset is important for better pose detection, clothing alignment, segmentation, and testing.

CREATED A FOLDER FOR DATASET:

STRUCTURE:

dataset/

|— men/

|— women/

|— jackets/

|— tshirts/

|— hoodies/

|— front_pose/

|— side_pose/

|— different_body_types/

Then,

Downloaded multiple images of each folder. For example, men and women in different different poses and attires. And kept them in their respective folder.

Remember to add unique fashion cloths and different poses so the model can be trained to be more accurate.

DATA SET CLEANING:

Dataset cleaning is the process of organizing, filtering, and improving collected data by removing poor-quality, incorrect, duplicate, or unsuitable images that may reduce the performance of an AI system. In a Virtual Try-On system, dataset cleaning is important because inaccurate or low-quality images can affect pose detection, body segmentation, clothing alignment, and try-on realism.

In this task, dataset cleaning was performed to improve the quality and usability of collected person and clothing images. Images with blurry quality, incomplete body visibility, multiple people, poor lighting, or incorrect clothing angles were removed. Only clear and properly aligned images were retained for testing and experimentation.

Delete images with:

- cropped body
- blurry image
- multiple people
- dark lighting
- side-hidden clothes
- weird background

Keep:

- full body visible
- front-facing clothing
- clean pose
- simple background

DATASET DOCUMENTATION:

Dataset documentation is the process of recording and organizing information about the collected dataset, including folder structure, image categories, dataset purpose, and usage. Proper documentation helps maintain clarity and improve dataset management.

In this task, dataset documentation was created to organize person images, clothing images, body types, and pose variations used for Virtual Try-On experiments.

DOCUMENTATION

Dataset Name: Virtual Try-On Dataset

Purpose:

This dataset is created for Virtual Try-On system testing and optimization. It is used for body landmark detection, segmentation, clothing alignment, pose testing, and try-on experiments.

Folder Description:*persons/*

Contains full-body person images used for pose detection and try-on testing.

poses/

Contains images with different body positions such as front pose, side pose, and hand movement for testing model robustness.

clothes/

Contains clothing images categorized by type such as jackets, t-shirts, hoodies, shirts, and pants.

segmented_output/

Contains processed images after background removal or segmentation.

test_images/

Contains images used for try-on experiments and testing.

workflow_docs/

Contains reports, workflow notes, dataset documentation, and challenge analysis.

Dataset Usage:

- Pose estimation
- Body segmentation
- Clothing alignment
- Virtual try-on testing
- AI optimization experiments

CONCLUSION:

In this task, I focused on dataset expansion and cleaning to improve the quality and organization of data used for the Virtual Try-On system. Different person images, clothing categories, body types, and pose variations were collected and arranged into a structured dataset for easier testing and experimentation. Dataset cleaning was also performed by removing poor-quality, blurry, incomplete, or unsuitable images to improve consistency and usability.